

Conclusion

Conclusion I

template_autoresearch_project shows how a tractable AutoResearch task can be represented as reproducible template infrastructure. The small MNIST neural-network classification experiment is small by design, but it exercises the method surfaces that matter for a public default: a declared research program, local input-data provenance, bounded candidate families (MLP, nearest-centroid, softmax regression, tiny patch-attention), objective scoring, budget and cost ledgers, evidence-linked claims, generated figures, benchmark grading, manuscript-variable hydration, loop-settlement ledgers, figure-quality checks, validation, local integrity attestation (passed), and human review gates.

Conclusion II

The broader field is converging across five related trends. Autoresearch systems seek end-to-end automation of ideation, experiment execution, writing, and review. Autoformalization pushes informal claims toward machine-checkable representations. ML-for-ML systems use search, evaluation, and evolutionary code loops to improve algorithms. Agentic science composes retrieval, planning, experimentation, and communication into higher-autonomy workflows. Structured knowledge synthesis supplies the substrate that all of those systems need: source-linked artifacts, knowledge graphs, tensors, vector spaces, and uncertainty-aware memories that can be navigated without losing provenance.

Conclusion III

This exemplar makes a deliberately narrower contribution. It does not mine the live literature, construct a Conceptual Nexus Model, search for Lean proofs, mutate arbitrary code, coordinate external agents, or approve its own paper. Instead it demonstrates governance infrastructure for automated research workflows: every important claim is hydrated from run artifacts, every generated figure is registry-bound and locally checked for source and pixel integrity, every budget and candidate decision is recorded, and review state remains outside generated self-approval. The security layer adds local inventory and checksum evidence while keeping external signing and production deployment claims out of scope.

Conclusion IV

The durable product is therefore not only the MNIST metric. It is a reproducible research process whose data, claims, captions, figures, review gates, and manuscript variables can be reviewed, rerun, and extended. As more ambitious automated-science systems mature, this kind of offline, deterministic, evidence-governed baseline remains useful because it preserves the part of scientific automation that should not be optional: inspectable provenance, explicit limits, and human-governed publication judgment.